

CSCE 222

Homework 3

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1 Section 2.4

1.1 Question 12 - Suppose that $A = \{2, 4, 6\}$, $B = \{2, 6\}$, $C = \{4, 6\}$, and $D = \{4, 6, 8\}$. Determine which of these sets are subsets of which other of these sets.

Definition: $A \subseteq B \iff \forall x(x \in A \rightarrow x \in B)$

A - None of the sets contain all elements of A, thus A is not a subset of B, C, or D

B - A is the only set that contains all elements of B, thus B is a subset of A

C - Only A and D contain all elements of C, thus C is a subset of A and D

D - None of the sets contain all elements of D, thus D is not a subset of A,B, or C

1.2 Question 14 - For each of the sets in Exercise 13, determine whether $\{2\}$ is an element of that set.

Definition: $A \subseteq B \iff \forall x(x \in A \rightarrow x \in B)$

1.3 a - $\{x \in R \mid x \text{ is an integer greater than } 1\}$

Because all elements of $\{2\}$ are real numbers and integers greater than 1, $\{2\}$ is an element of that set.

1.4 **b - $\{x \in R \mid x \text{ is the square of an integer}\}$**

There exists an element, 2, in $\{2\}$ that is not the square of an integer as $\sqrt{2} \notin \mathbb{Z}$.

Thus $\{2\}$ is not an element of that set.

1.5 **c- $\{2, \{2\}\}$**

There exists an element in $\{2, \{2\}\}$ that is equivalent to $\{2\}$, thus $\{2\}$ is an element of that set by definition.

1.6 **d- $\{\{2\}, \{\{2\}\}\}$**

There exists an element in $\{\{2\}, \{\{2\}\}\}$ that is equivalent to $\{2\}$, thus $\{2\}$ is an element of that set by definition.

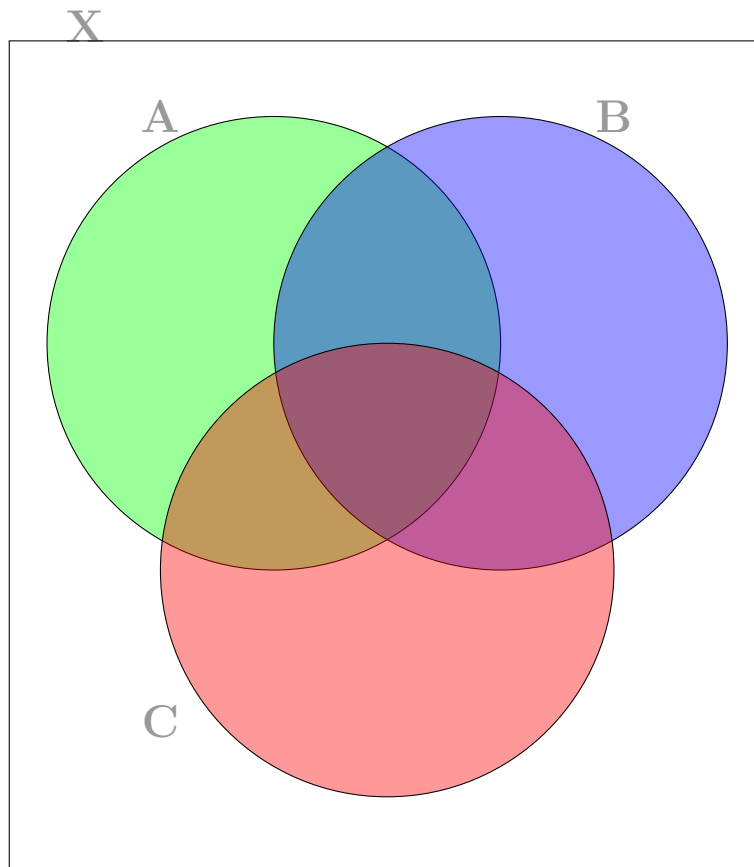
1.7 **e- $\{\{2\}, \{2, \{2\}\}\}$**

There exists an element in $\{\{2\}, \{2, \{2\}\}\}$ that is equivalent to $\{2\}$, thus $\{2\}$ is an element of that set by definition.

1.8 **f- $\{\{\{2\}\}\}$**

By exhaustion, there does not exist an element in $\{\{\{2\}\}\}$ that is equivalent to $\{2\}$, thus $\{2\}$ is not an element of that set by definition.

1.9 Question 22a - Use a Venn diagram to illustrate the relationships $A \subset B$ and $A \subset C$



1.10 Question 22b

1.11 Question 32

2 Section 5.1

2.1 Question 8

2.2 Question 12

2.3 Question 14

2.4 Question 18